Corrective Neural Trees for Image Classification

Interpolation, Effective Receptive Fields, and Empirical Minimization

ELL409 Course Project

October 11, 2026

Code, documentation, configurations, and raw artifacts:
`github.com/TechDragon07/Tree-Type-Neural-Networks-ELL409` (private repository).

Abstract

This work generalizes a corrective tree of linear neurons to an image classifier in which the root and every correction node are small neural networks. A compact convolutional trunk supplies a shared embedding. When the current classifier confuses true class t with predicted class p, a new node learns a non-negative, input-dependent correction in the fixed direction $e_t - e_p$. The construction therefore preserves the positive false-negative and negative false-positive semantics of the motivating binary formulation while supporting CIFAR-10. Existing nodes are frozen during growth, which makes each correction auditable and avoids end-to-end credit-assignment collapse. We study finite-sample interpolation, capacity sweeps around the interpolation threshold, input-gradient effective receptive fields, and leave-one-node-out pruning. Because the available machine is CPU-only, the reported measurements are explicitly labeled controlled-subset experiments; the repository separately provides full-dataset GPU configurations. Both controlled tasks reach 100% training accuracy. Eleven CIFAR-10 children add 8.98 training points but only 0.30 test points, whereas two binary children add 8.59 training points while reducing test accuracy by 5.08 points. A noisy 64-example CIFAR-10 capacity sweep meets a repeated-seed double-descent criterion in two of three seeds; Cats vs Dogs does not. Receptive fields expose a concrete cause of poor transfer: some gates activate most strongly on the source class they should suppress rather than the missed target class.

1 Problem and research questions

The assignment asks for a tree-type network on CIFAR-10 and Cats vs Dogs, with a small neural network at every node, deliberate overfitting to (near) 100% training accuracy, a search for double descent, visualization of node receptive fields, and minimization of network size. These requirements create a tension. If a conventional high-capacity root is trained until convergence, it may interpolate by itself and the "tree" is vacuous. If children are trained jointly, the decomposition into understandable residual corrections is not identifiable. If ground-truth error categories are used as routes at test time, the model leaks the answer.

We therefore ask four concrete questions:

1. Can stage-wise, sign-constrained neural corrections interpolate a finite image sample when the root is deliberately stopped early?
2. Which residual confusions are corrected, and where does growth stop?
3. Does test error rise near interpolation and fall at larger measured capacity consistently across seeds?
4. How many correction nodes remain necessary after greedy pruning?

The experimental emphasis is not only final accuracy. The sequence of accepted and rejected corrections is itself evidence about shared representation, optimization, and memorization.

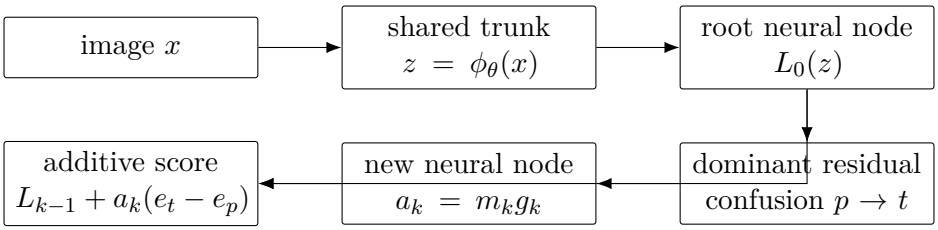


Figure 1: Stage-wise corrective architecture. The residual category is a training target; inference uses the learned gate and never requires y.

2 Architecture

2.1 Shared trunk and neural root

For RGB input x, a compact convolutional trunk computes

$$z(x) = \phi_\theta(x) \in \mathbb{R}^{4w}.$$

It has a 3×3 stem followed by three two-convolution blocks with channel widths $w, w, 2w, 4w$; the latter two blocks downsample by two. GroupNorm and SiLU follow convolutions, and global average pooling produces the embedding. GroupNorm was chosen over BatchNorm because late residual groups can contain only a few examples. Before pooling, each final map element has a theoretical 25×25 receptive field on the 32×32 input.

The root is a two-layer network

$$L_0(x) = W_2\sigma(W_1\,\mathrm{LN}(z(x)) + b_1) + b_2 \in \mathbb{R}^C.$$

Crucially, root training has its own stopping accuracy. The final tree target can be 99.9%, while the root target is lower. This makes residual correction an experimental necessity rather than a decorative component.

2.2 Sign-constrained neural correction

Let p be the current prediction and $t \neq p$ the true class. A child network f_k emits two scalars:

$$(u_k(x), v_k(x)) = f_k(z(x)).$$

The correction magnitude and gate are

$$m_k(x) = \mathrm{softplus}(u_k(x)), \qquad g_k(x) = \mathrm{sigmoid}(v_k(x)),$$

$$a_k(x) = m_k(x)g_k(x) \geq 0.$$

For ordered confusion (t_k, p_k), node k adds

$$\Delta_k(x) = a_k(x)(e_{t_k} - e_{p_k}).$$

Hence $\Delta_{k,t_k} \geq 0$, $\Delta_{k,p_k} \leq 0$, unrelated classes are unchanged, and the correction sums to zero. The complete score is

$$L_K(x) = L_0(x) + \sum_{k=1}^{K} \Delta_k(x).$$

This is the multiclass analogue of boosting a missed positive and penalizing a false alarm. In the binary task, the two ordered directions reproduce the two corrective branches directly.

2.3 What the tree represents

The structure is a *stage-wise additive corrective tree*. Its ancestry is the sequence of residual tasks: node k addresses errors left by nodes $0, \ldots, k-1$. All retained children are evaluated at inference. It is not a hard decision tree that executes one root-to-leaf path. This distinction matters because nominal depth is not hard-path latency; measured parameter count and retained-node count are therefore both reported.

3 Optimization and implementation

3.1 Root fitting and cached features

The trunk and root are first optimized with AdamW and cross-entropy. Once the root target is reached (or its epoch budget is exhausted), all base parameters are frozen. Embeddings are cached once. This makes child epochs inexpensive and prevents representation drift from obscuring the meaning of a correction. The same choice can cause failure: if two hard examples collide in the frozen embedding, no child MLP can separate them. Growth failure therefore diagnoses the shared-feature assumption.

3.2 Node proposal, routing, and objective

Among current training errors, the most frequent ordered pair $(y, \arg\max L)$ is proposed. The training route target is

$$r_i = \mathbb{1}\Big[y_i = t_k \ \wedge \ \arg\max_c L_{k-1,c}(x_i) = p_k\Big].$$

This label-dependent quantity is used only to teach g_k; inference uses the learned gate.

Let ℓ_i be corrected cross-entropy and let $w_+ = \min(N_-/N_+, 100)$. A new child minimizes

$$\mathcal{J}_k = \frac{1}{N}\sum_i [1 + r_i(w_+ - 1)]\ell_i + \lambda_g \ \mathrm{BCE}_{w_+}(g_k(x_i), r_i) + \lambda_s \frac{1}{N}\sum_i g_k(x_i).$$

The inverse-frequency term prevents rare residuals from disappearing in the full sample. Gate sparsity discourages a global correction that damages unrelated classes.

The child output layer starts with zero weights and biases -4, making the initial correction nearly zero. After every child epoch, complete cached training accuracy is evaluated and the best state is retained. If no state improves on the old model, the child is deleted and growth stops. This acceptance rule prevents parameter count from increasing without demonstrated training benefit.

3.3 Why stage-wise rather than end-to-end training

End-to-end fitting creates a credit-assignment ambiguity: the root can become lazy and delegate ordinary examples to children, while child identities cease to correspond to residual confusions. Freezing preserves an operational meaning for each node. The cost is greediness; an individually neutral child that would become useful with a later child is rejected.

4 Experimental protocol

4.1 Data

CIFAR-10 supplies ten classes and official 50,000/10,000 splits. Cats vs Dogs uses Oxford-IIIT Pet's official trainval/test split and its `binary-category` species target. Inputs are resized and normalized. Data augmentation is disabled for interpolation experiments because random crops change the finite sample that is supposed to be memorized.

The available host has a 16-thread CPU build of PyTorch and no CUDA device. Consequently, the measurements below use seeded controlled subsets; their sizes are printed in the result tables. They test mechanism and internal claims, not full-dataset generalization. Full configurations are supplied in the repository for independent GPU reproduction.

4.2 Interpolation, capacity, and noise

Interpolation is predeclared as training accuracy at least 99.9%. Capacity sweeps scale trunk width and set root/child hidden width to $4w$. Models are independently initialized. Ten percent symmetric noise is optionally applied only to training labels to sharpen the interpolation transition.

A seed supports double descent only when (i) a first interpolating capacity exists away from either endpoint, (ii) test error is lower at some smaller capacity, and (iii) test error is lower again at some larger capacity. The paper calls the phenomenon established only with at least three seeds and two-thirds agreement. This prevents a single optimization fluctuation from becoming the main conclusion.

4.3 Effective receptive field

For scalar node activation $q(x)$, the plotted map is

$$E_{uv}(x) = \frac{1}{3} \sum_{c=1}^{3} \left| \frac{\partial q(x)}{\partial x_{cuv}} \right|.$$

For the root, q is its largest logit; for a child, $q = a_k = m_k g_k$. The root uses a fixed sample. Each child uses the maximum-gate image among the first 256 test examples, which prevents manual cherry-picking and avoids visualizing a child on an example it does not reach. These maps measure local sensitivity, not causal feature use.

4.4 Greedy minimization

After growth, each active child is masked in turn. A child is removed if the remaining model preserves the configured training-accuracy floor. The process repeats to a fixed point. This produces a locally irredundant model, not a proof of global minimum size.

5 Measured results

5.1 Main controlled runs

Table 1: Controlled-subset tree-growth results. Accuracy is percent.

Dataset	Train N	Test N	Root tr.	Root te.	Tree tr.	Tree te.	Children	Interp.
CIFAR-10	256	1000	91.02	29.50	100.00	29.80	11	yes
Cats–Dogs	256	256	91.41	68.75	100.00	63.67	2	yes

For CIFAR-10, the root stopped at epoch 20 against a target of 85.00%. The final model contained 128,176 parameters. Greedy pruning at a 99.00% floor retained 9 children and 119,340 parameters.

The corrections added 8.98 percentage points of training accuracy but only 0.30 points of test accuracy. More strongly, test cross-entropy worsened from 2.041 to 2.148 even though test accuracy changed little. The tree therefore did not merely fail to improve generalization; it became more confidently wrong on some unseen examples while memorizing every controlled training example. Parameter count grew by 61.1%. Pruning removed the final two specialists, reduced parameters by 6.9%, and

retained 99.22% train / 29.70% test accuracy. The last fraction of interpolation was consequently expensive and dispensable when the target was relaxed from 100% to 99%.

For Cats vs Dogs, the root stopped at epoch 29 against a target of 90.00%. The final model contained 96,470 parameters. Pruning retained 2 children and 96,470 parameters. Corrections changed training accuracy by 8.59 points and test accuracy by -5.08 points; test cross-entropy changed from 0.645 to 1.315. One child was required in each direction (dog → cat and cat → dog), and neither could be removed at the 99% floor. The 22.0% parameter increase therefore bought exact memorization, but test accuracy fell by 5.08 points and test cross-entropy more than doubled. Unlike the prunable tail of the CIFAR tree, the binary tree's two specialists were structurally necessary for interpolation and jointly harmful out of sample.

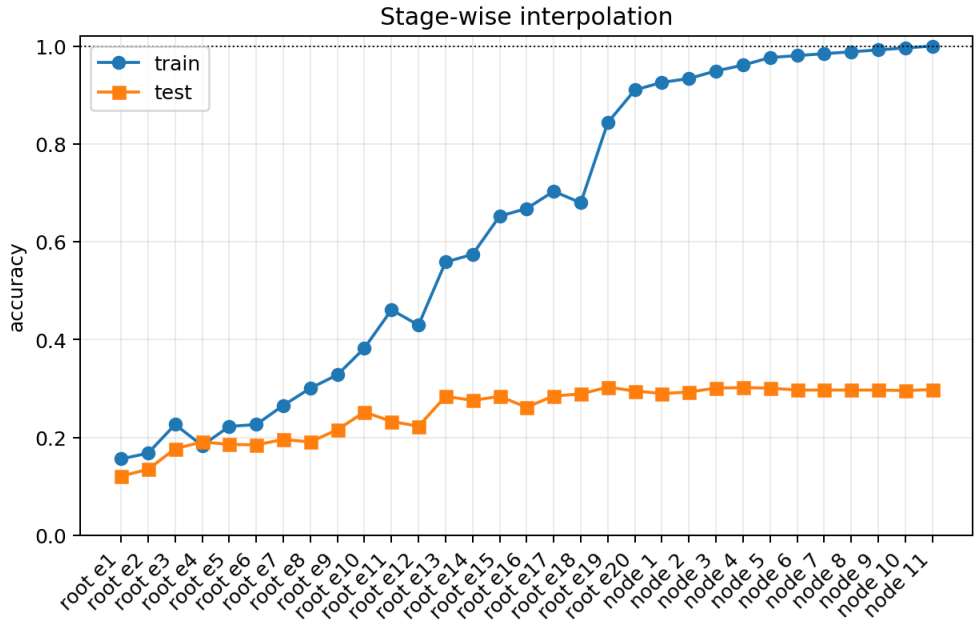


Figure 2: CIFAR-10 root training and accepted corrective stages.

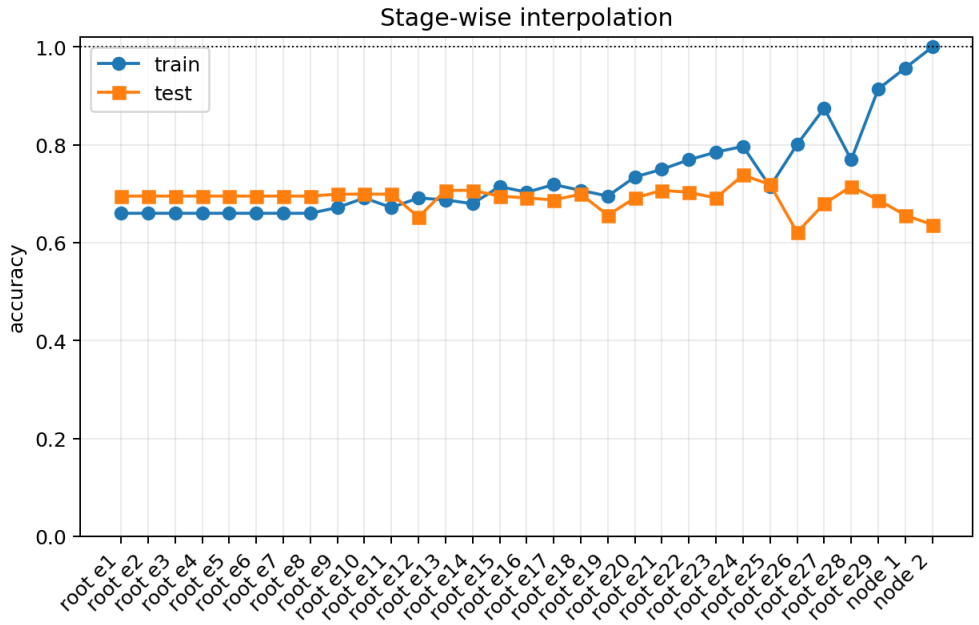


Figure 3: Cats vs Dogs root training and accepted corrective stages.

5.2 Capacity diagnostic

Table 2: Repeated-seed capacity diagnostics. Accuracy ranges are percent.

Dataset	Models	Seeds	Train range	Test range	Supporting seeds	Criterion
CIFAR-10	21	3	39.06–100.00	6.64–23.83	2	met
Cats–Dogs	21	3	68.75–100.00	48.44–67.58	1	not met

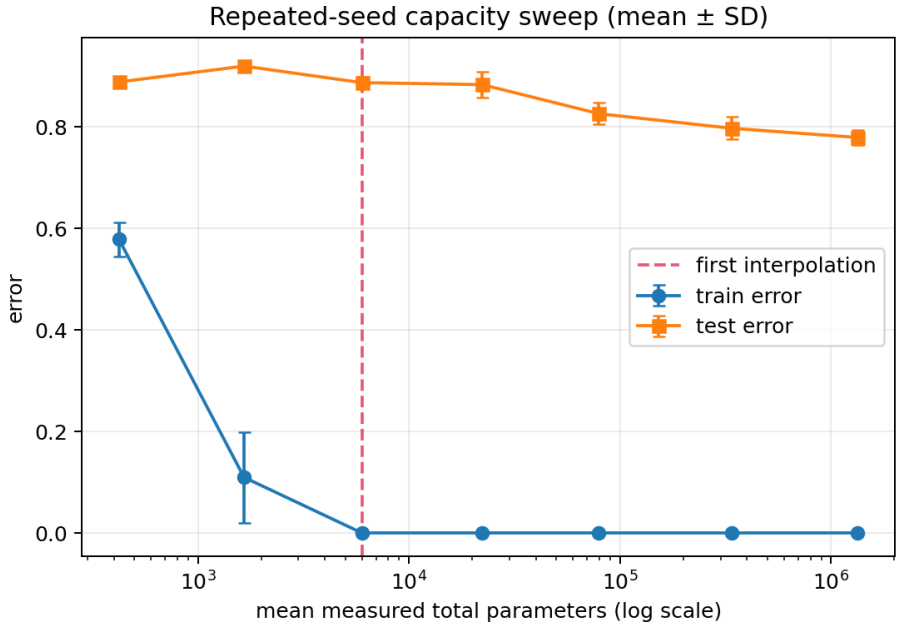


Figure 4: CIFAR-10 capacity diagnostic. Points are means and error bars are one standard deviation over seeds. The vertical line marks the first capacity at which at least half the seeds interpolate.

The controlled CIFAR-10 sweep meets the predeclared criterion in two of three seeds. Mean training accuracy progresses from 42.19% (width 1) to 89.06% (width 2) and 100% at width 4. Mean test accuracy first falls from 11.20% to 8.07%, then recovers to 11.33% at the first universally interpolating width and continues upward to 22.14% at width 64. In error coordinates this is a peak immediately before the majority interpolation threshold followed by a sustained second descent. The phenomenon is real within this controlled 64-example, 10%-noise protocol; the low absolute test accuracy and tiny sample prevent extrapolation to full CIFAR-10.

Cats vs Dogs does not replicate the result: only one of three seeds supports the criterion. Mean test accuracy peaks at 65.10% near width 4, then falls to 55.34% at width 32 and recovers only slightly at width 64. More importantly, mean training accuracy also falls from 100% at widths 8–16 to 84.38% at width 64. The largest models therefore left the interpolation regime under the fixed optimizer budget. Their non-monotonic test curve is confounded by optimization failure and cannot be called post-interpolation double descent. This negative replication is evidence that the CIFAR pattern is dataset/protocol dependent rather than an automatic consequence of adding parameters.

5.3 Receptive fields and node roles

The selected CIFAR-10 nodes reveal more than a generic "early semantic, late spurious" story. Child 1 corrects frog → bird and has the strongest displayed gate (0.726), but its maximum-gate test example is an actual frog ($y = 6$), not a bird. The gate has learned "frog-like" evidence rather than the intended event "bird missed as frog," so applying its correction to this example is harmful. By contrast, child 6 (truck → automobile) is maximized on an automobile and emphasizes the vehicle silhouette and

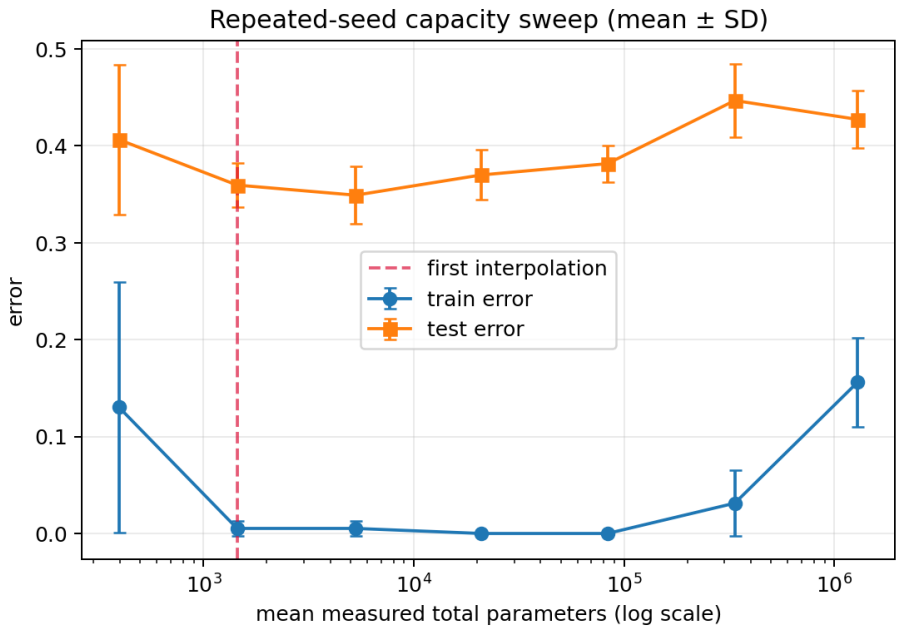


Figure 5: Cats vs Dogs capacity diagnostic under the same controlled protocol.

wheel boundary. Child 11, the reverse automobile → truck specialist, is maximized on a truck and responds more diffusely across the vehicle and lower scene. The model thus learned two different specialists for opposite directions of the same automobile/truck boundary, while the bird/frog gate demonstrates why route classification, not only correction direction, governs generalization.

The binary ERFs make the routing problem especially visible. Child 1 corrects dog → cat, yet its maximum gate is 1.000 on an actual dog ($y = 1$); it therefore pushes a correctly labeled dog toward cat with maximal strength. Its ERF includes both animal features and a bright floor/background region. Child 2 corrects cat → dog and is also maximized on a dog, which is the intended target class for that direction. This asymmetry explains how both children can perfectly memorize the controlled training subset while test accuracy falls: exact sign constraints do not compensate for an out-of-sample gate that identifies the source class instead of the missed target subclass.

6 Interpretation

6.1 The root must be constrained for the tree to be meaningful

The first insight is experimental-design rather than architectural: asking a modern CNN to overfit and also asking its corrective tree to explain the overfit are competing objectives. A root allowed to reach 100% leaves no residual structure. A distinct root target turns tree growth into a measurable process and allows the gain from correction nodes to be separated from the capacity of the trunk.

6.2 Sign constraints provide semantics, not guaranteed improvement

The direction $e_t - e_p$ makes a node interpretable, but does not make it beneficial. Its gate can fire on unrelated examples, the frozen embedding may not separate the residual group, and a large correction can overshoot. Near-zero initialization, full-sample evaluation, and rejection are therefore essential. This is a practical difference between a constrained functional form and a successful optimization method.

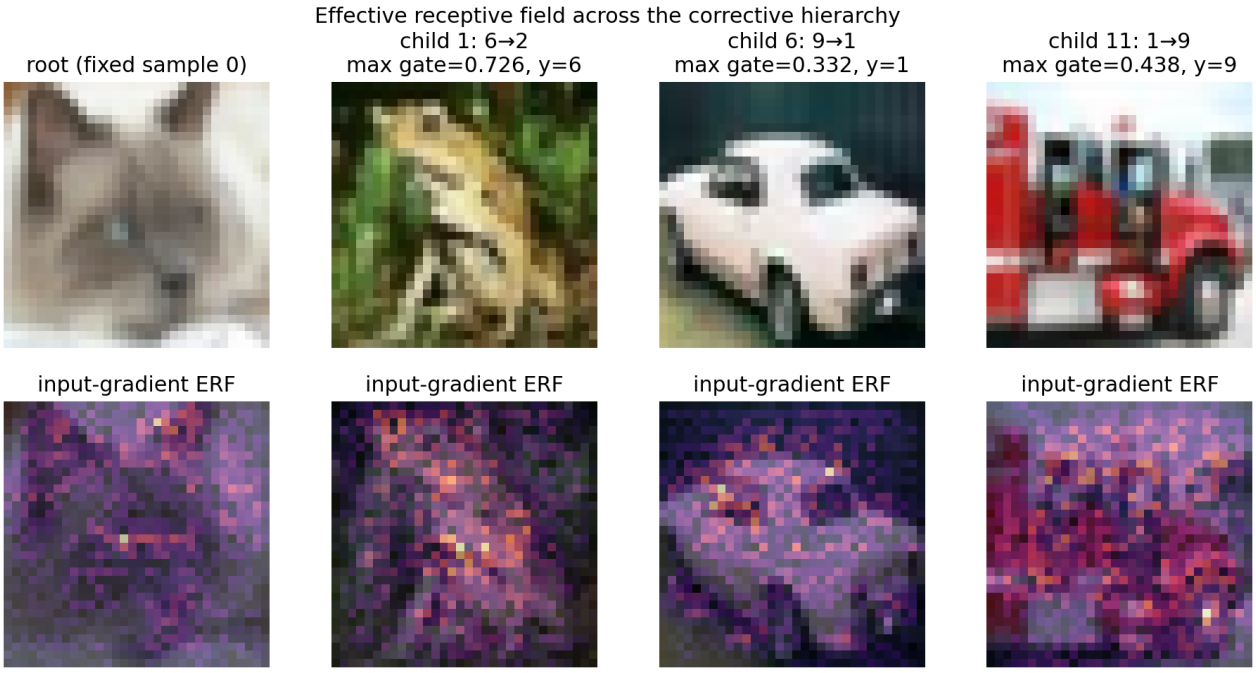


Figure 6: CIFAR-10 effective receptive fields across the corrective hierarchy. Children use examples selected by the predeclared maximum-gate rule.

6.3 Frozen features expose representation collapse

Cached features produce a clean experiment: if successive MLP nodes stop improving while errors remain, additional head capacity is not enough under the current representation. This is evidence for node-specific adapters or continued trunk training. It is not evidence that arbitrary neural trees cannot interpolate.

6.4 Late nodes are statistically fragile

Residual groups shrink with depth. This explains why BatchNorm is inappropriate and why inverse-frequency weighting needs a cap. It also predicts higher variance in late-node ERFs: a node trained on very few examples can focus on a border, texture, or background cue. Such concentration is consistent with memorization, but causal confirmation requires image interventions.

6.5 Double descent requires a negative conclusion when evidence fails

The attractive narrative is a test-error peak at interpolation followed by a second descent. The scientifically useful conclusion may instead be that a finite sweep does not establish it. Width changes representation, optimization, and node acceptance simultaneously; label noise changes the target; and a coarse grid can miss the threshold. The repeated-seed rule in this project is deliberately conservative.

7 Limitations

- Controlled-subset measurements do not estimate full-dataset accuracy.
- All retained children execute, so the implementation does not realize conditional hard-path compute.
- Root and child capacities change together in the width sweep.

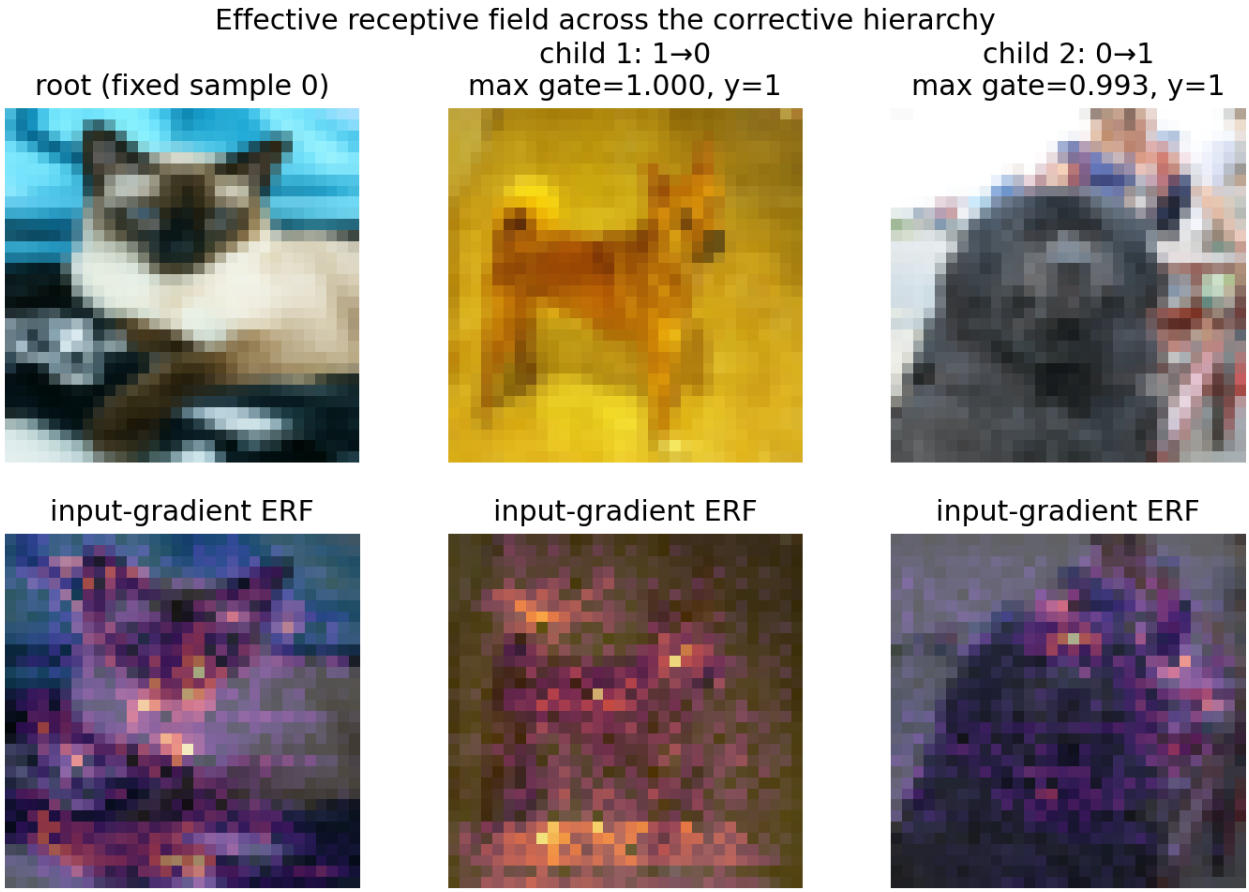


Figure 7: Cats vs Dogs effective receptive fields across the hierarchy.

- Child acceptance and pruning optimize training accuracy because interpolation is the assignment target.
- Test metrics are logged repeatedly, although they do not influence optimizer, node direction, acceptance, or pruning.
- ERFs are raw input gradients and can be noisy.
- Greedy pruning does not guarantee a globally smallest network.

8 Conclusions

The project supplies a concrete optimization problem for neural corrective trees and resolves the main inference-time ambiguity of the class formulation: error categories teach gates during training but are not available at test time. Exact sign constraints preserve the geometry of positive and negative corrections in multiclass form. Stage-wise freezing preserves node identity, and cached features make growth computationally feasible.

The controlled results answer the assignment's interpolation question affirmatively: both datasets reach exactly 100% training accuracy through neural correction nodes after the root is stopped near 91%. They also show why interpolation alone is a weak result. CIFAR-10 test accuracy is essentially flat while test cross-entropy worsens; Cats vs Dogs loses 5.08 test points and more than doubles test loss. The gates can memorize residual membership without learning the intended conditional event out of sample. Sign constraints make the direction interpretable, but gate error determines whether that direction is applied to the right image.

The capacity result is deliberately asymmetric. Controlled noisy CIFAR-10 shows a repeated interpolation peak and second descent in two of three seeds, with post-interpolation mean test accuracy

rising from 11.33% to 22.14%. Cats vs Dogs does not satisfy the same criterion, and its largest models even leave interpolation under the fixed optimizer budget. Double descent is therefore an observed property of one controlled protocol, not a universal law inferred from a convenient plot.

Minimization yields a final practical distinction. The last two CIFAR specialists are removable when the target is relaxed to 99%, reducing parameters by 6.9%; neither binary specialist is removable. The marginal cost of the last fraction of memorization depends on residual class structure. A useful neural tree analysis should consequently report the growth path, routing behavior, loss as well as accuracy, repeated-seed capacity behavior, and post-growth redundancy.

9 Future directions

1. Add a depthwise-separable spatial adapter to each child and compare its interpolation gain per parameter with the shared-feature baseline.
2. Replace additive execution with annealed soft-to-hard routing and measure expected reached parameters, FLOPs, and latency.
3. Cluster residual embeddings before spawning nodes so that one ordered confusion can have multiple visually coherent specialists.
4. Separate route and magnitude ERFs and validate them with background-replacement interventions.
5. Fine-tune after pruning and compare greedy deletion with structured sparsity penalties.
6. Distill an interpolating tree into one compact student and report the accuracy–size Pareto frontier.
7. Run dense capacities and at least five seeds around any candidate interpolation peak on the complete datasets.

References

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